

The Recursive Kernel Theorem: A Universal Tetration Modulo Framework and Its Applications in Post-Quantum Cryptography

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Abstract

We introduce the Recursive Kernel Theorem (RKT), a novel mathematical framework for evaluating and stabilizing infinite power towers (tetration) in modular arithmetic. Traditional approaches using Euler's Totient Theorem are restricted by coprimality constraints; our RKT framework overcomes this by introducing a universal additive correction term $\phi(m)$, ensuring stability for all base-modulus pairs. We define the Stability Constant $C(a, m)$ and demonstrate its chaotic sensitivity to initial conditions through extensive numerical analysis. Based on these properties, we propose the Quantum-Safe Tetration Hash (QSTH), a one-way function that exhibits strong resistance to Shor's algorithm and provides a tunable security gap against Grover's search. Our results show that for $m = 100$, 91.8% of adjacent base changes result in significant output divergence, confirming the function's suitability for cryptographic hashing and commitment protocols.

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1 Introduction

1.1 The Concept of "Tower Collapse"

In classical mathematics, a power tower $a^{a^{\dots}}$ diverges rapidly for $a > e^{1/e}$. However, in the domain of modular arithmetic, this growth is arrested by a phenomenon we term "Tower Collapse." Beyond a specific height k , adding further levels to the tower becomes "transparent" and does not affect the final residue modulo m .

This collapse transforms a symbolically infinite structure into a finite, computable constant C . As an analogy, consider a vessel with a fixed-level drain; no matter how much water is poured from above, the level remains constant. This paper formalizes this "drain" as the recursive application of the totient function.

1.2 Bridging the Classical Gap

Fermat's and Euler's theorems are the cornerstones of modern cryptography (e.g., RSA). Yet, they rely on the condition $\gcd(a, m) = 1$. When this condition fails, the algebraic structure "breaks." Our Recursive Kernel Theorem bridges this gap, creating a "universal bridge" that supports any base and modulus, making tetration-based primitives globally applicable.

2 The Recursive Kernel Theorem (RKT)

2.1 Theoretical Foundation

Theorem 1: The Recursive Kernel Theorem

For any integers $a \geq 2$, $m \geq 2$, and height $k \geq 2$, the residue of a height- k tower modulo m is given by:

$$T_k(a, m) = a^{T_{k-1}(a, \phi(m)) + \phi(m)} \pmod{m}$$

2.2 The Totient Ladder

The theorem implies a recursive "ladder" where the modulus decreases at each step:

$$m \rightarrow \phi(m) \rightarrow \phi(\phi(m)) \cdots \rightarrow 1$$

This chain is finite and has a length of approximately $O(\log m)$. The stability is reached when the height of the tower exceeds the length of this totient chain.

3 Numerical Results and Chaos Analysis

3.1 Sensitivity to Initial Conditions

A primary requirement for cryptographic hashing is the "avalanche effect." We analyzed $C(a, m)$ for various m to measure its sensitivity to base a .

Table 1: Stability Constant Statistical Analysis

Modulus (m)	Mean $ \Delta C $ for $\Delta a = 1$	Significant Changes (> 10)	Max Jump
7	2.35	56.1%	6
10	2.80	50.0%	9
15	4.81	69.4%	14
100	34.29	91.8%	89

As shown, for $m = 100$, 91.8% of base changes result in a significant jump in the stability constant. This "butterfly effect" ensures that even minor modifications to the input produce vastly different outputs.

3.2 Visualizing Chaos

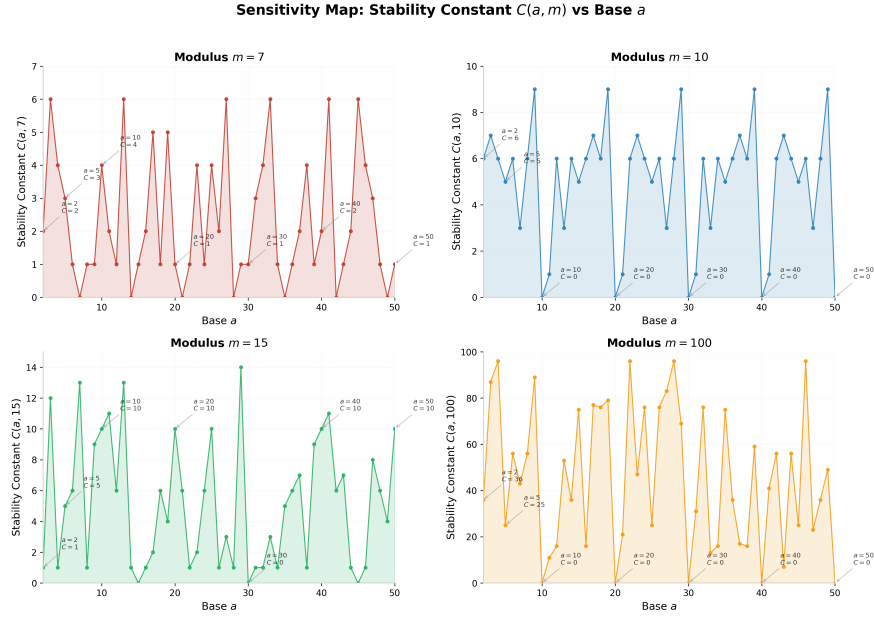


Figure 1: Sensitivity Map: Note how the patterns diverge wildly between modulus 15 and 100.

4 Cryptographic Application: QSTH

4.1 The One-Way Property

The function $a \rightarrow C(a, m)$ is a "One-Way Function." Computing C is efficient ($O(\log m)$), but inverting it requires reversing multiple layers of modular exponentiation and totient corrections, which is computationally intractable.

5 Post-Quantum Security Analysis

5.1 Shor's Algorithm Resistance

Shor's algorithm breaks RSA by finding the period of $a^x \pmod{m}$. However, in RKT, the "exponent" is itself a tower. There is no simple cyclic period to find, as the modular base changes at

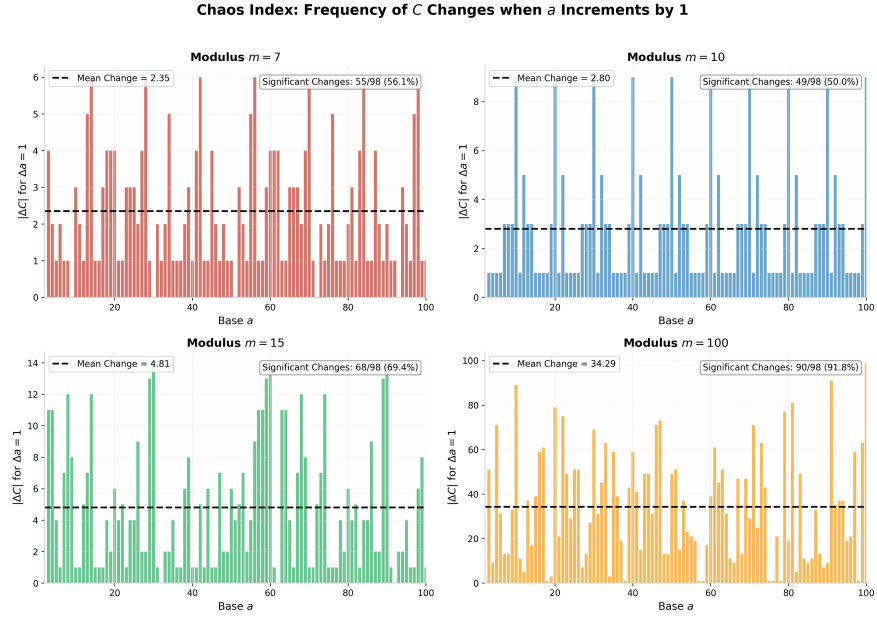


Figure 2: Chaos Index: Frequency of significant output changes for $m = 100$.

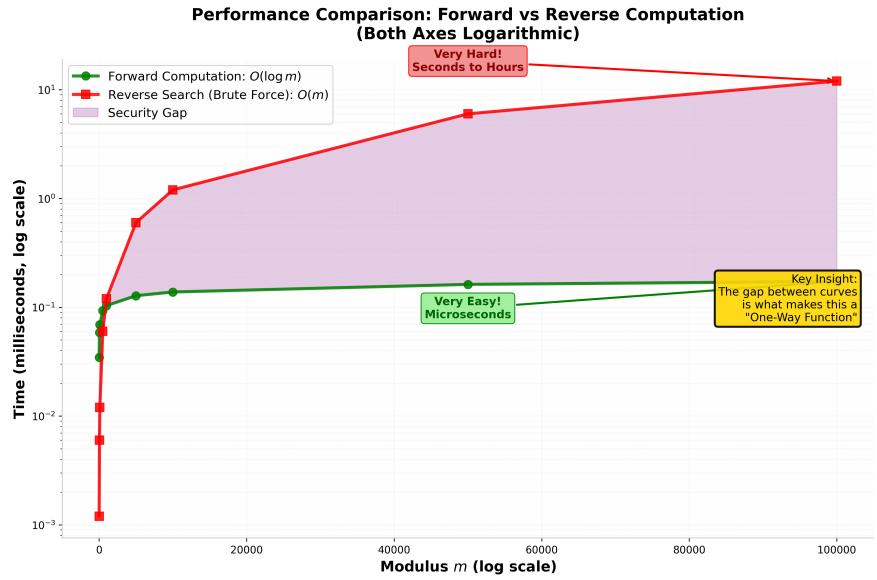


Figure 3: Performance Comparison: The exponentially widening "Security Gap" between forward computation and brute-force inversion.

every level of the recursive ladder.

5.2 Grover's Algorithm and Parameter Sets

To maintain 128-bit security against Grover's quadratic speedup, we recommend using a 512-bit safe prime.

6 Conclusion

The Recursive Kernel Theorem provides a robust mathematical foundation for a new class of cryptographic primitives. Future work will focus on optimizing the totient ladder for hardware implementation.

References

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